

Md. Arman Hossen

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EDUCATION

PhD in Physics

August 15, 2024

Washington University in St. Louis, Missouri, USA.

Dissertation: “Development of a Balloon Mission to Test Flight a Novel Mini-Dilution Refrigerator and a TES Gamma-Ray Detector Array, and New Measurement of the Spin of the Black Hole Cygnus X-1”

MS in Physics

2015

Jahangirnagar University, Bangladesh.

Dissertation: “Nonlinear Structures in Degenerate Astrophysical Compact Objects”, ISBN (978-3-659-79687-6)

BS in Physics

2014

Jahangirnagar University, Bangladesh.

ACADEMIC AND RESEARCH APPOINTMENTS

Postdoctoral Researcher at Penn State University

March 15, 2024 - Present

- Lead calibration, integration, and performance validation of Hybrid CMOS X-ray detector systems for the NASA-funded BlackCAT CubeSat mission, designed to detect high-energy transients such as gamma-ray bursts and provide rapid triggers to ground-based observatories.
- Develop and operate advanced detector test facilities, including high-vacuum systems, liquid-nitrogen cooling systems, and Python-based PID temperature control loops for precision detector characterization.
- Design automated Python pipelines for instrument control, data acquisition, signal processing, spectral fitting, and statistical uncertainty propagation.
- Conduct detailed detector characterization, including quantum efficiency, gain calibration, read noise analysis, energy resolution, and event grading for next-generation space-based X-ray instrumentation.
- Collaborate with multidisciplinary teams of scientists and engineers on detector development, systems integration, calibration campaigns, and flight readiness activities for future NASA missions.

PhD researcher at Washington University in St. Louis

2018-February 2024

- Developed and characterized Transition-Edge Sensor (TES) gamma-ray microcalorimeters for the NASA-supported DR-TES stratospheric balloon mission, requiring cryogenic integration, superconducting detector operation, and precision calibration.
- Contributed to the X-Calibur and XL-Calibur hard X-ray polarimetry balloon missions through calibration and integration of CdZnTe (CZT) detector systems and associated readout electronics.
- Built data acquisition and analysis pipelines to process detector signals, optimize performance, and quantify instrument response using advanced statistical and signal-processing techniques.
- Performed detector calibration, cryogenic, electronics, and vacuum subsystems to improve system reliability and scientific performance. Supported NASA-funded instrumentation projects that successfully flew scientific payloads and produced peer-reviewed publications.

MS researcher

2014-2015

Modeling of the nonlinear plasma systems, i.e., electron-ion plasma, dusty plasma, etc., mostly present in compact objects like neutron stars, pulsars, etc.

Publications and Impact (Status as of May 2026)

Total articles 20+

Conference Proceedings 15+

h-Index 12 (Google Scholar)

Total Citations 292 (Google Scholar)

RECENT PUBLICATIONS and MANUSCRIPTS IN PREPARATION*

1. **M. A. Hossen et al., .. et al.**, "Quantum Efficiency Measurements of the Speedster-EXD550 hybrid CMOS detector across the soft X-ray band", J. Astron. Telesc. Instrum. Syst., (under publication).
2. **M. A. Hossen et al.**, Impact of the Inclination of Cygnus X-1 on the Estimated Black Hole Spin, in preparation.
3. W. Bevidas, .. **M. A. Hossen .. et al.**, "Characterization of random telegraph noise in an H2RG X-ray hybrid CMOS detector," J. Astron. Telesc. Instrum. Syst. 11(2) 026002 (7 April 2025) <https://doi.org/10.1117/1.JATIS.11.2.026002>
4. F. Shirazi, E. Gau, **M. A. Hossen, et al.**, "511-CAM mission: a pointed 511 keV gamma-ray telescope with a focal plane detector made of stacked transition edge sensor microcalorimeter arrays," J. Astron. Telesc. Instrum. Syst. 9(2), 024006 (2023).
5. F. Kislat, ...**M. A. Hossen ...et al.**, "ASCENT: a balloon-borne hard X-ray imaging spectroscopy telescope using transition-edge sensor microcalorimeter detectors," J. Astron. Telesc. Instrum. Syst. 9(1), 014002 (2023).
6. Q. Abarr, ... **M. A. Hossen .. et al.**, "Performance of the X-Calibur hard X-ray polarimetry mission during its 2018/19 long-duration balloon flight", Astropart. Phys. 143, 102749 (2022).

EARLIER PUBLICATIONS

1. **M. A. Hossen** and A. A. Mamun, Solitary and shock structures in a strongly coupled cryogenic quantum plasma, **Physics of Plasma (USA)** **22**, 073505 (2015).
1. **M. A. Hossen** and A. A. Mamun, Nonlinear ion-acoustic waves in a degenerate plasma with nuclei of heavy elements, **Physics of Plasma (USA)** **22**, 102710 (2015).
2. **M. A. Hossen** and A. A. Mamun, Nonlinear positron-acoustic waves in fully relativistic degenerate plasmas, **Physica Scripta (Sweden)** 91, 035201 (2016).
3. **M. A. Hossen** and A. A. Mamun, "Nonlinear Electroacoustic Waves in Fully Relativistic Astrophysical Degenerate Plasmas," in *IEEE Transactions on Plasma Science*, vol. 44, no. 4, pp. 643-650, April 2016, doi: 10.1109/TPS.2016.2530838.
4. **M. A. Hossen et al.**, Heavy-ion-acoustic solitary and shock waves in an adiabatic multi-ion plasma, **Brazilian Journal of Physics (Brazil)** 45, 444 (2015).
5. **M. A. Hossen et al.**, Modified ion-acoustic shock waves and double layers in a degenerate electron-positron-ion plasma in the presence of heavy negative ions, **Brazilian Journal of Physics (Brazil)** 44, 703 (2014).
6. **M. A. Hossen et al.**, Study of the higher-order shock excitations in a degenerate quantum plasma, **Journal of the Korean Physical Society (Korea)** 65, 1883 (2014).
7. **M. A. Hossen et al.**, Instability Analysis of Positron-Acoustic Waves in a Magnetized Multi-Species Plasma, *Commun. Theor. Phys.* 67 458 (2017).
8. M. R. Hossen, **M. A. Hossen**, S. Sultana, and A. A. Mamun, Modeling of modified ion-acoustic shock waves in a relativistic electron degenerate multi-ion plasma for higher order nonlinearity, **Astrophysics and Space Science (Netherlands)** **357**, 34 (2015).
9. P. R. Dip, **M. A. Hossen**, M. Salahuddin, and A. A. Mamun, Shock waves and double layers in a

- quantum electron-positron-ion plasma, **Journal of the Korean Physical Society** (Korea) 68, 520 (2016).
10. P. R. Dip, **M. A. Hossen**, M Salahuddin, A. A. Mamun, Cylindrical and spherical electro-acoustic waves in a strongly coupled cryogenic quantum plasma, *J. High Temp.*, Volume 58, 781-788. (2020).
 11. M. M. Hasan, **M. A. Hossen**, A. A. Mamun, Acoustic solitary waves in a magnetized degenerate quantum plasma, *Physics of Plasmas*, Volume 24, Issue 7 (2017).
 12. P. R. Dip, **M. A. Hossen**, M Salahuddin, A. A. Mamun, Effects of Bohm potentials and Fermi temperatures on nonplanar solitary and shock excitations in a strongly-coupled quantum plasma, *J. of the Korean Physical Soc.*, Vol. 70, Pages 777-784, (2017).
 13. M. M. Hasan, **M. A. Hossen et al.**, Effect of Bohm quantum potential in the propagation of ion-acoustic waves in degenerate plasmas, *Chinese Phys. B* **25** 105203 (2016).

CONFERENCES PROCEEDINGS

1. Development and initial characterization of next-generation event-driven hybrid CMOS X-ray detectors, T. R. Emeigh *et al.*, *Proc. SPIE* (2026), submitted.
2. Source centroiding and angular resolution with BlackCAT and beyond, L. L. O'Neill *et al.*, *Proc. SPIE* (2026), submitted.
3. A low-noise small-pixel hybrid CMOS detector for future X-ray observatories, L. R. Stone *et al.*, *Proc. SPIE* (2026), submitted.
4. Flight software and trigger algorithms for the BlackCAT CubeSAT coded aperture telescope, K. Nizam *et al.*, *Proc. SPIE* (2025), in press.
5. Status of testing and characterization of a low-noise small-pixel hybrid CMOS detector, L. R. Stone *et al.*, *Proc. SPIE* (2025), in press.
6. Fine-Resolution Low-Noise X-Ray Hybrid CMOS Detectors For Future High-Throughput X-Ray Missions, L. R. Stone *et al.*, *AAS/High Energy Astrophysics Division 57* (5), 303.03, (2025).
7. Quantum efficiency measurements of the Speedster-EXD550 hybrid CMOS detectors at the Mn K α and K β energies, M. A. Hossen *et al.*, *Proc. SPIE, UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* 13625, 261-271 (2025).
8. The BlackCAT CubeSat: a soft X-ray coded aperture telescope, A. D. Falcone *et al.*, *Proc. SPIE, UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* 13625, 136250E (2025).
9. Mechanical design and implementation of the BlackCAT payload, M. Wages *et al.*, *Proc. SPIE, 13625, UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* (2025).
10. Detection of random telegraph noise in hybrid CMOS detectors, W. A. Bevidas, *et al.*, *Proc. SPIE, UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV*, 13625, (2025).
11. Calibration measurements for the BlackCAT CubeSat x-ray coded aperture telescope, T. R. Emeigh *et al.*, *Proc. SPIE, UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* 13625, 517-524 (2025).
12. Status of x-ray hybrid CMOS detector development efforts at PSU, "Status of x-ray hybrid CMOS detector development efforts at PSU," L. R. Stone *et al.*, *Proc. SPIE* 13103, X-Ray, Optical, and Infrared Detectors for Astronomy XI, 131030M (2024).
13. BlackCAT: an upcoming soft X-ray coded aperture telescope on a 6U CubeSat, A. D. Falcone *et al.*, *Proc. SPIE* 13093, *Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray*, 1309317 (2024).
14. Ground calibration of the BlackCAT CubeSat x-ray Coded Aperture Telescope, J. M. Colosimo *et al.*, *Proc. SPIE* 13093, *Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray*, 1309376 (2024).

15. Characterization of the x-ray telescope after the first flight of XL-Calibur, H. Kuramoto *et al.*, Proc. SPIE 12679, Optics for EUV, X-Ray, and Gamma-Ray Astronomy XI, 126791B (2023).

TALKS IN CONFERENCES

1. Test of a Mini-Dilution Refrigerator and a Transition Edge Sensor Array on a Stratospheric Balloon, **M. A. Hossen** *et al.*, 2023 Scientific Ballooning Technologies Workshop, May 17-19, (2023).
2. Transition-edge sensor microcalorimeter detectors for future X-ray and gamma-ray mission, **M. A. Hossen** *et al.*, AAS#240, Bulletin of the American Astronomical Society, Vol. 54, No. 6, (2022).
3. Plans for the Test of a NIST Microcalorimeter Array for Hard X-ray and γ -ray Astronomy on a Stratospheric Balloon Flight, **M. A. Hossen** *et al.*, 19th International Workshop on Low-Temperature Detectors, 2021.
4. Tests of NIST TES Sensor Arrays with Collimated X-ray Beams, **M. A. Hossen** *et al.*, AAS#237, 14th Jan (2021).
5. NIST microcalorimeter arrays for the hard X-ray and γ -ray astronomy, **M. A. Hossen** *et al.*, Proc. SPIE. 11838, Hard X-Ray, Gamma-Ray, and Neutron Detector Physics XXIII, (2021).
6. Implementing and investigating the numerical accuracy of the exact Compton scattering in the x-ray reflection from the black hole accretion disk, **M. A. Hossen** *et al.*, American Astronomical Society Meeting Abstracts, 53 (1), 404.08 (2021).
7. Investigating the Linear Polarization of X-ray Binary Sources with XL-Calibur, L. Lisalda..**M. A. Hossen** *et al.*, AAS High Energy Astrophysics Division meeting #20, id. 103.41. Bulletin of the American Astronomical Society, Vol. 55, No. 4 (2023).
8. The Performance of DR-TES (Dilution Refrigerator Transition Edge Sensor) Mission, K. Hu, **M. A. Hossen** *et al.*, AAS/High Energy Astrophysics Division, (2024).
9. The 511-CAM Gamma Ray Mission Concept, H. Krawczynski .. **M. A. Hossen** *et al.*, AAS/High Energy Astrophysics Division 21, 105.33 (2024).
10. DR-TES and 511-CAM missions: transition-edge sensor microcalorimeter detectors for pointed X-ray and Gamma-ray missions, N. R. Caveno, ..**M. A. Hossen** *et al.*, Proc. SPIE PC13103, X-Ray, Optical, and Infrared Detectors for Astronomy XI, PC1310309 (2024).
11. A hard X-ray polarimetry view of the Crab Nebula and Pulsar from the balloon-borne telescope XL-Calibur, E. Gau, ..**M. A. Hossen** *et al.*, 245th Meeting of the American Astronomical Society, id. 341.08. Bulletin of the American Astronomical Society, Vol. 57, No. 2 (2025).

POSTER

1. Optimization of a Precision Pointing System for Calibration of the BlackCAT CubeSAT, Brynn Bortree *et al.*, SPSCon. (2025).

RESEARCH EXPERTISE AND SKILLS

- X-ray Instrumentation, i.e., X-ray and Gamma-ray detectors calibration for astrophysical applications, TESs, cryogenics.
- X-ray beamline operation, i.e., at Penn State, our lab uses a 47-meter beamline for calibrating X-ray detectors and a coded mask aperture for astrophysics missions.
- High-energy Astrophysics.
- Software integration and X-ray/gamma-ray detector data analysis with Python and C++.
- Astrophysical software: root, XSPEC, and Sherpa.
- 30-hour Physics machine shop training course

TEACHING AND MENTORING EXPERIENCE

- Lecturer, Introductory Physics (Engineering Students), Daffodil International University (2017)
- Teaching Assistant, Introductory Physics Labs, Washington University (2018–2019)
- Physics Outreach Coordinator, Washington University (2022–2023)
- Graduate Research Co-supervisor (2 MS students; 5+ co-authored publications)
- Member, Department Diversity Committee, Washington University (past), Penn State (current)
- Postdoc peer mentor, Penn State

Outreach Activities

Conducted physics demonstrations and interactive sessions for audiences of various age groups, promoting scientific curiosity and public engagement in STEM.

ORIGINAL SCIENTIFIC CONTRIBUTION

Worked as an active member for the NASA mission X-Calibur (2018), XL-Calibur (2024), DR-TES (2024), and BlackCAT (current).

EDITORIAL, REFEREEING & SERVICE EXPERIENCE

- High Energy Density Physics (Elsevier)
- Zeitschrift für Naturforschung A (Germany)
- Journal of Applied Mathematics and Physics
- Multiscale and Multidisciplinary Modeling, Experiments, and Design
- Familiarity with the Editorial peer-review processes and publishing system

AWARD

National Science & Technology (N.S.T.) fellowship (2014-2015), the Ministry of Science and Technology (MOST), Bangladesh.

PROFESSIONAL MEMBERSHIP

American Physical Society (APS), SPIE membership, National Postdoctoral Association (NPA)

*(complete list of publications is available in Google Scholar)

REFERENCES

Prof. Abe Falcone Postdoctoral Advisor Penn State

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Prof. Suvrath Mahadevan Associate Head, Postdoctoral Affairs Penn State

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Prof. Randall McEntaffer Department Head Penn State

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